Paleomagnetism of Réunion Island and Its Bearing on Secular Variation

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The results of a paleomagnetic study of the volcanics of Réunion Island are reported. The samples fall into three distinct age groups (0–0.6 m.y., 1 m.y., and 2.0 m.y.), and the mean directions for these groups are in reasonable agreement with the values expected for a geocentric axial dipole field. An analysis of the group 1 data shows that the secular variation is consistent with the present day nondipole variation for the latitude of Réunion. If comparable sampling densities are assumed, a comparison of the Réunion and Hawaiian paleomagnetic data suggests that the present day asymmetry between the northern and the southern hemisphere, in the behavior of he nondipole components, has persisted throughout the past 600,000 years and that the dipole wobble has a time constant greater than this value.

INTRODUCTION

In recent years volcanic oceanic islands have been the subject of intensive geological and geophysical studies because of their increasing importance in many geophysical hypotheses. Studies have been made on islands in the Pacific Ocean, in particular the Hawaiian Islands, but little work has as yet been done on the islands in the Indian Ocean. The Mascarene Islands, Réunion, Mauritius, and Rodriguez, appeared to resemble the Hawaiian Islands in several respects; consequently, field work on these islands was undertaken by Ian McDougall in July 1963 and a large number of orientated samples were collected for paleomagnetic work. This communication is the first of a series in which the paleomagnetic results from the Mascarene Islands will be reported.

The islands are paleomagnetically of interest for two principal reasons. First, the thick sequences of lava flows are ideal for studying the frequencies of polarity reversals of the geomagnetic field by combining radiometric age determinations with paleomagnetic polarity measurements on the same rock units. The results of such a study on the Réunion material have been reported [*Chamalaun and McDougall*, 1966], and they showed that polarity reversals are indeed worldwide and contemporaneous, thus leaving little doubt as to the reality of the field reversal hypothesis. Second, analyses of the present geomagnetic field by *Cox* [1962] and *Creer* [1962] have shown that there are marked differences between the Pacific region and its antipodal region, which includes the Mascarene Islands, in the behavior of the nondipole as well as similar differences between the northern and southern hemispheres. A detailed paleomagnetic study by *Doell and Cox* [1965] on the main island of Hawaii suggested that these differences may have persisted for at least a million years. In this paper the paleomagnetic results for Réunion will be compared with the Hawaiian data, and their significance with regard to the problem of secular variation will be discussed.

GEOLOGY OF RÉUNION

Réunion, situated at the southwest extremity of the Mauritius-Seychelles ridge (55°33′E, 21°S), consists of two volcanos, Piton des Neiges, an extinct and deeply eroded volcano, and Piton de la Fournaise, which is still active. The geology has been described by a number of authors, but the most recent account is given by *Upton and Wadsworth* [1965, 1966], whose nomenclature will be followed here. Figure 1 is a geological sketch map of the island.

Piton des Neiges, occupying the northwestern two-thirds of the island, has an over-all shield shape and is deeply dissected by gorges that have exposed the older formations. Amphitheater-headed valley erosion has resulted in the excavation of huge cirques in the central part

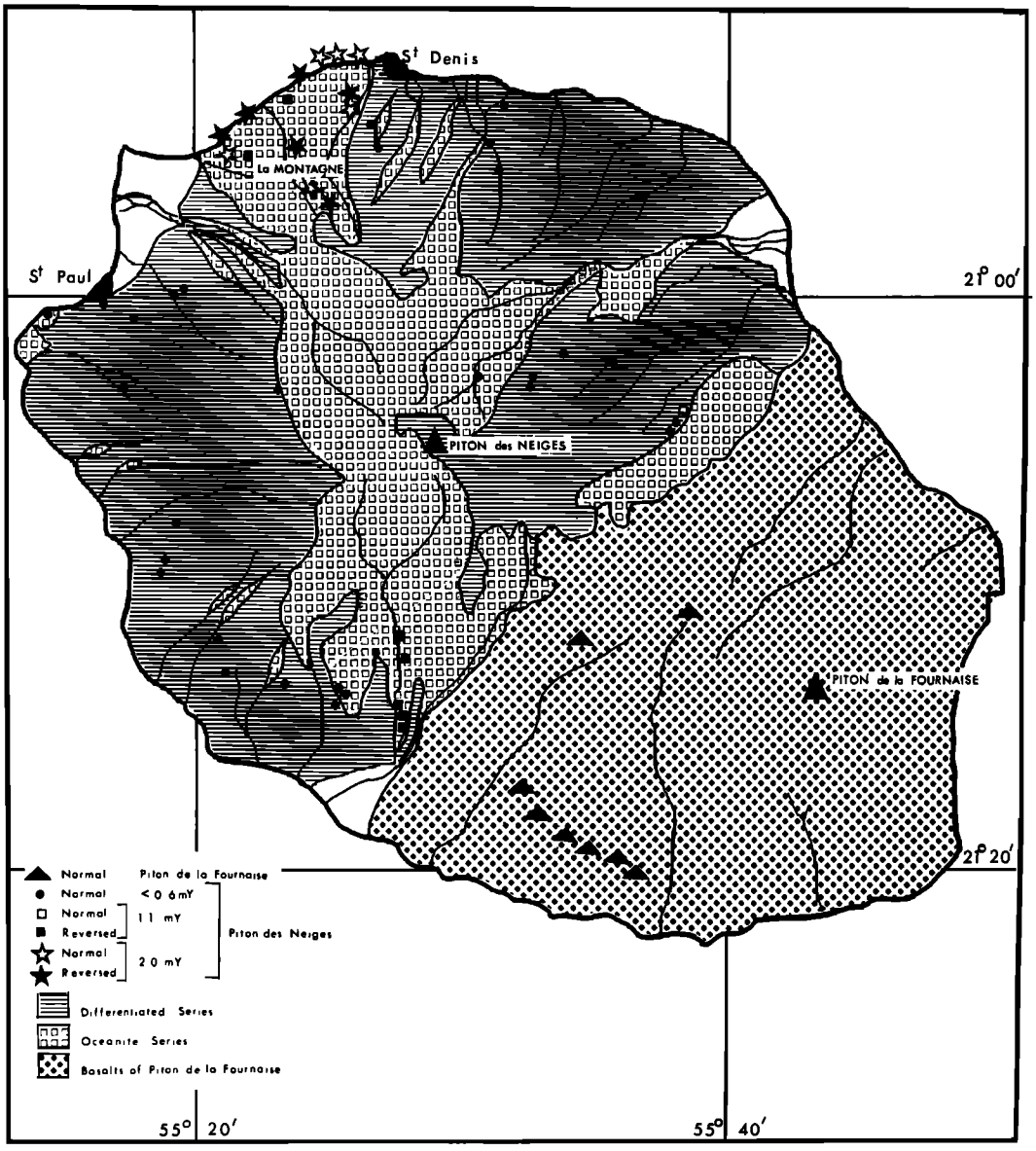


Fig. 1. Simplified geological map of Réunion Island showing sample locations [after *Upton and Wadsworth*, 1965].

of the volcano. The volcanic sequences of Piton des Neiges have been divided into two major groups: an older series consisting mainly of olivine-rich basalts and termed the oceanite series and a younger series consisting of intermediate lavas and feldsparphyric basalts called the differentiate series. The differentiated series forms a surface veneer over most of the western and eastern flanks of Piton des Neiges. In general, the flows of the differentiated series tend to be thicker than the flows of the oceanite series. The oceanite series are most exposed in the gorges, but they also form a number of relatively upstanding blocks, the largest being the massif of La Montagne.

Piton de la Fournaise covers the southeastern part of the island and consists of olivine basalts grading into oceanites. Petrologically,

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the basalts resemble the oceanite series of Piton des Neiges.

Radiometric dating [*Chamalaun and McDougall*, 1966] has shown that Piton des Neiges was active at least 2.0 m.y. ago and that the change from oceanite to differentiated series occurred about 0.5 m.y. ago. Piton de la Fournaise was active at least during the same period as the formation of the differentiated series.

Experiment

More than one hundred samples were collected from unweathered and well exposed outcrops on both volcanos. The first collection was made by Ian McDougall, and later additional samples were collected by M. McElhinny, in particular, from the La Montagne massif. The samples were orientated by means of a Brunton compass and their localities are shown in Figure 1. Initially the NRM directions of two specimens per sample were measured. On the basis of these measurements, thirty representative specimens were selected for stepwise alternating-field demagnetization to 600 oe and thermal demagnetization to 700°C, using the instruments and methods described by *Irving et al.* [1961*a*, *b*]. The af demagnetization results for the test specimens showed that an alternating field of 150 oe removed the unstable components, while sufficient intensity was retained to allow measuring on a short-period astatic magnetometer. Subsequently two specimens from each sample were treated with this field, and these 'cleaned' results are used in this paper. (The flow average data after af demagnetization are given in the appendix.)

Compared with the Hawaiian rocks, for which typically 50–70% of the intensity still remains at 150 oe, the Réunion rocks are less stable in that only 30–50% of the intensity remains at that field. Typical af curves (Figure 2) show the rapid decrease in intensity with field indicating that most of the remanence is associated with low coercivities.

The thermal demagnetization curves (Figure 2*a*) display a variety of shapes, and low blocking temperatures are predominant. They also suggest that more than one magnetic phase is present. To test this possibility, a number of specimens were measured on an automatic recording balance. Some examples are shown in Figure 2. The saturation magnetization decreases rapidly with temperature without producing a well defined Curie point, and the cooling curve is quite different from the heating curve suggesting alteration of the magnetic phase. The results indicate that the magnetic mineral is a titanomagnetite with a wide range of composition. Further detailed experimental work is now in progress to study the magnetochemistry of these rocks.

In Figure 3 normal and lognormal histograms are plotted for the intensity and susceptibility. A lognormal distribution fits the data rather well, as was previously determined by *Irving et al.* [1966] and *Tarling* [1966]. No significant differences were found between normal and reversely magnetized rocks or between rocks of different ages. The mean intensity and susceptibility are 3.6×10^{-3} emu/cm^3 and 0.2×10^{-3}, respectively, which agree with the values assumed by *Vine and Mathews* [1963] for the interpretation of the magnetic anomalies associated with seamounts in the Indian Ocean.

Paleomagnetic Results

The directions of magnetization fall essentially into two groups, parallel and antiparallel to the present field, representing normal and reversed polarities as discussed previously [*Chamalaun and McDougall*, 1966]. In the absence of dated samples standard paleomagnetic practice would have involved computing a mean pole for each of these two groups, assuming that each group comprised enough samples to have averaged out the secular variation. In the present case, however, a relatively large number of samples have been dated radiometrically. Figure 4 is a summary of the polarity versus age data, which shows that there are three distinct age groups present. The first group comprises samples from Piton de la Fournaise and the differentiated series of Piton des Neiges, with ages ranging from the present to 0.6 m.y. All samples have normal polarity and belong therefore to the Bruhnes normal epoch. The second group contains samples of the oceanite series of Piton des Neiges with ages between 1.0 and 1.3 m.y. Both polarities are present since the samples lie on either side of the boundary between the Jarramillo normal event and the Matuyama reversed epoch.

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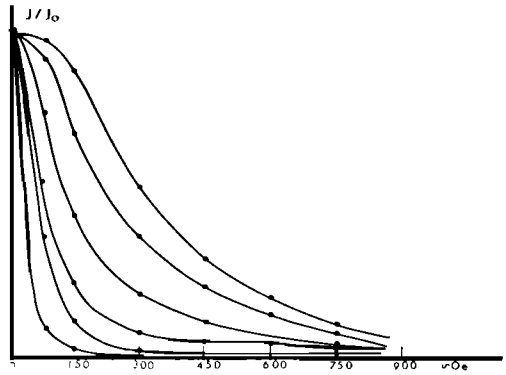


Fig. 2*a*.

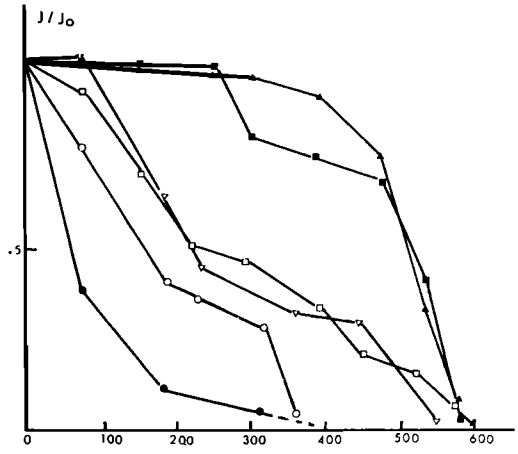


Fig. 2*b*.

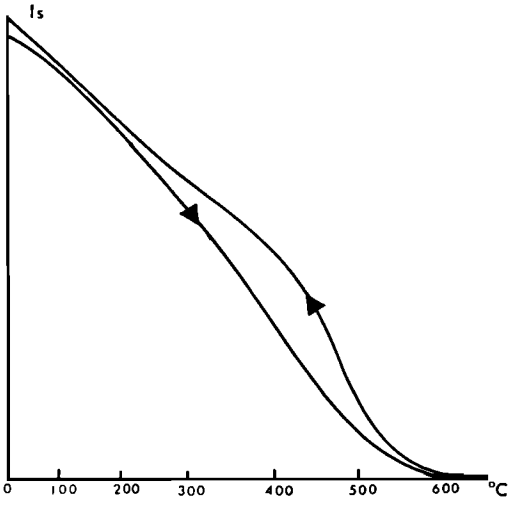


Fig. 2*c*.

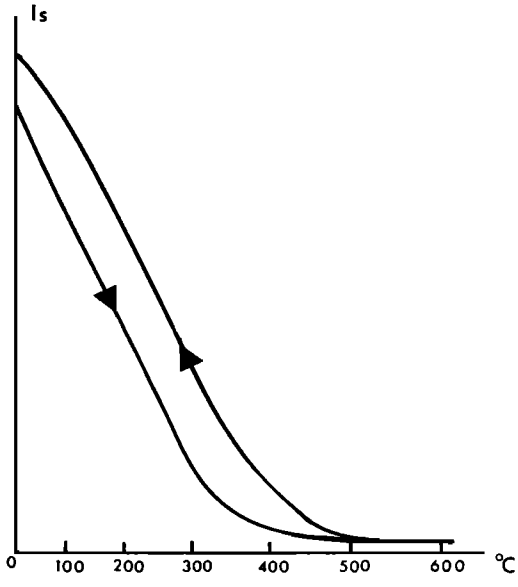


Fig. 2*d*.

Fig. 2. Shows examples of typical af demagnetization curves (*a*), thermal demagnetization curves (*b*), and saturation magnetization versus temperature curves in an applied field of 5000 oe (*c* and *d*).

The third group consists of samples from the oceanite series of Piton des Neiges collected from the La Montagne massif. Normal and reversed flows were found with ages agreeing closely at 2.0 ± 0.1 m.y. The normal polarity rocks belong to the Olduvai normal event and the reversed ones to the Matuyama reversed epoch. At one locality rocks with reversed polarity conformably overlie rocks with normal polarity.

It is not certain whether these three age groups represent three distinctly different phases of volcanic activity, or whether they are the result of some peculiar sampling bias. The dated samples were, however, selected on the basis of minimum alteration, by regarding their polarity and such to give a reasonable coverage of the age range as deduced from the field relationships. These dated samples are therefore thought to be representative of the

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whole collection, and so all the samples should fall within these three age groups. The difficulty is, however, to allocate the undated samples to their correct age group when definite stratigraphical control is lacking. In a volcanic terrain it is possible for adjacent flows to be greatly different in age. The following may serve as an example to illustrate the difficulty.

In the La Montagne massif all the oceanite series samples, with the exception of two, were dated at about 2.0 m.y.; one would therefore be inclined to allocate all the undated samples to group 3. The two exceptions were reversely magnetized samples dated at 1.1 m.y. and belong to group 2. It is therefore possible that several other reversely magnetized samples now allocated to group 3 in fact belong to group 2. In making the allocation, however, the petrology, field relationships, and polarity were taken into account as completely as possible, and it is thought that few samples, if any, are actually misplaced.

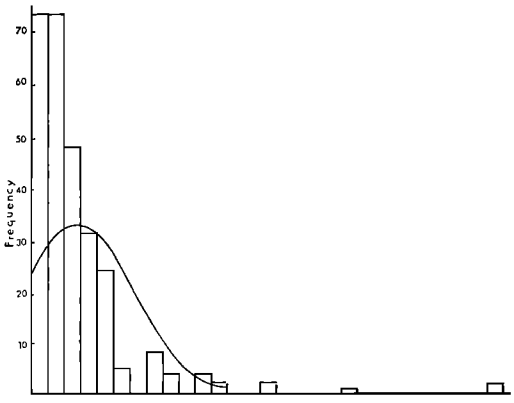


Fig. 3*a*. Normal histogram of the intensity of NRM. Arithmetic mean equals 3.6×10^{-6} emu/cm^3; class width, 1.27×10^{-3} emu/cm^3; standard deviation, 4.38×10^{-3} emu/cm^3; chi square, 2.24×10^{11} (df = 28).

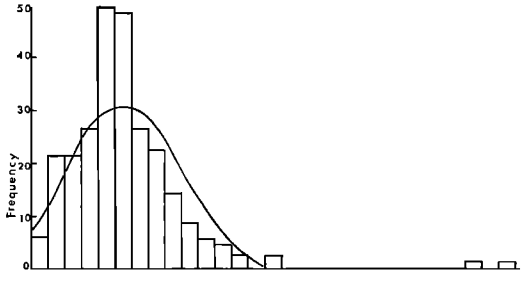


Fig. 3*b*. Normal histogram of susceptibility. Arithmetic mean equals 0.26×10^{-3} emu/cm^3/oe; chi square, 8.2×10^8 (df = 28).



Fig. 3*c*. Lognormal (base *e*) histogram of the intensity of NRM. Mean log equals 11.53 (equals 3.1×10^{-3} emu/cm^3); class width, 0.35; standard deviation, 0.72; chi square, 87.2 (df = 23).

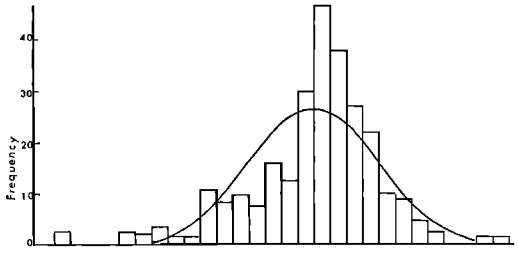


Fig. 3*d*. Lognormal (base *e*) histogram of susceptibility. Mean log equals 7.70 (equals 0.22×10^{-3} emu/cm^3/oe); class width, 1.60; standard deviation, 4.45; chi square, 561.7 (df = 28).

The uneven distribution of the samples in time makes it doubtful whether the computation of an over-all mean is justified. Therefore, in Table 1 the results are given in terms of the three groups with a further subdivision according to the polarity. In Figure 5 stereographic projections are shown of the directions of magnetization with respect to the present horizontal. Unit weight has been given to flows irrespective of the number of samples collected in each. The error circles [*Fisher*, 1953] overlap one another, indicating that the four means may not be significantly different. This overlapping can be interpreted as evidence for the internal consistency of the data and as evidence that the 'cleaning' procedure has been fairly successful. The same conclusion can be reached from considering the fact that the normal and reversed polarities are almost exactly antiparallel. In this particular case it so happens that the over-all mean is unaffected by the way in which the samples could be combined.

Discussion

When compared with the Hawaiian results

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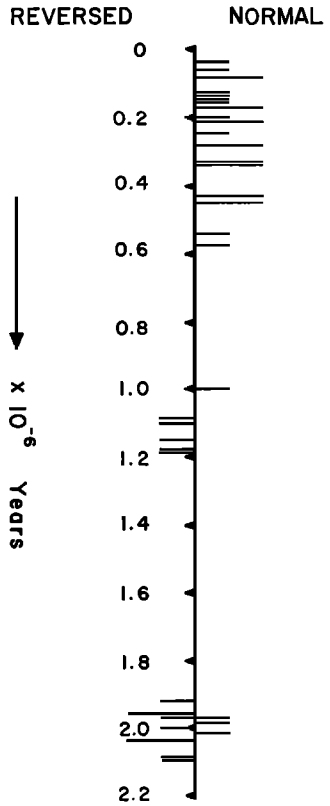


Fig. 4. Plot of polarity versus age for individual flows. Short lines indicate one flow; long lines, two flows.

[*Doell and Cox*, 1965], the Réunion directions are more widely scattered.

Experimental errors and intrinsic variations of the geomagnetic field are the two main factors contributing to the dispersion of a set of magnetization vectors. The intrinsic variations are of considerable geophysical interest. By using sequences of well dated young rocks we may be able to obtain more information about the time constants that govern them. In this respect the Réunion data are unique, because for the first time such a set of samples with well defined age limits is available.

The contribution of experimental errors is thought to be small. Orientation errors could amount to a few degrees, but, where multiple samples from the same flow were obtained, the internal agreement was found to be very good. Imperfect cleaning might be another cause of scatter, but specimens that deviated by a large angle from what appeared to be the main group were further cleaned without obtaining a significant improvement. Furthermore, the fact that the normal and reversed direction are almost exactly antiparallel suggests that the cleaning procedure has been reasonably successful. In some samples a distinct flow pattern was found, which suggested the possibility of errors due to shape anisotropy, but neither high-field [*Stacey*, 1960] and low-field [*King and Rees*, 1962] torque meter experiments nor the acquisition of TRM in the laboratory suggested that magnetic anisotropy would be responsible for serious errors. *Doell and Cox* [1965] discuss at length the possible effects

TABLE 1. Summary of Paleomagnetic Data

		Mean of Flow Average Directions							Pole		Mean of Flow Average Poles					
		N	D, deg	I, deg	r	k	α_{95}, deg	δ_0	ϕ_1, deg	ψ_1, deg	ϕ_2, deg	ψ_2, deg	R	K	A_{95}, deg	Δ
Group 1 0.1–0.6 m.y.	Normal	49	006.3	−39.2	46.13	16.71	05.1	19.8	157.8	+83.9	184.3	+83.2	46.16	16.95	05.0	19.6
Group 2 1.0–1.3 m.y.	Normal	1	003.0	−35.8					124.7	+86.8						
	Reversed	9	191.3	+42.4	07.65	05.83	23.3	33.5	346.1	−78.9	313.9	−79.6	07.33	04.79	26.2	36.9
	Mean	10	190.1	+41.6	08.61	06.49	20.5	31.7	344.3	−80.1	345.3	−80.2	08.33	05.40	22.9	34.8
Group 3 1.9–2.0 m.y.	Normal	12	007.9	−37.0	11.15	12.94	12.5	22.5	144.1	+82.5	181.1	+81.0	11.25	14.61	11.7	21.1
	Reversed	26	178.5	+34.8	24.52	16.90	07.1	19.7	197.8	−87.7	162.3	−87.9	24.64	18.33	06.8	18.9
	Mean	38	181.1	+35.4	35.60	15.43	06.1	20.6	272.3	+88.2	352.1	−89.1	37.75	16.43	05.9	19.9

Notes

N, number of flows.
r, R, length of resultant vector.
k, K, Fisherian precision parameter.
δ_0, Δ, angular dispersion factor.
α_{95}, A_{95}, angle of confidence.

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that reheating may have on the directions of magnetization, but they discounted it as an important source of error both because of the magnetochemistry of the rocks and because of the field evidence. For Réunion data, there is insufficient evidence to evaluate this possibility fully. Another source of error is tectonic movement due to block faulting, caldera collapse, and gravity sliding, but Ian McDougall (personal communication) states that the field observations do not support such a suggestion except perhaps in the La Montagne sector. Although the experimental errors may not be altogether negligible, we will assume that their contribution to the scatter is small.

The intrinsic sources of scatter arise from time-dependent variations of the geomagnetic field itself caused by three main phenomena. First, the geomagnetic field is not strictly dipolar over the entire surface of the earth but is regionally perturbed by nondipole components attributed to local eddies in the core. Variations in the strength and positions of these eddies will appear as scatter of the paleo-

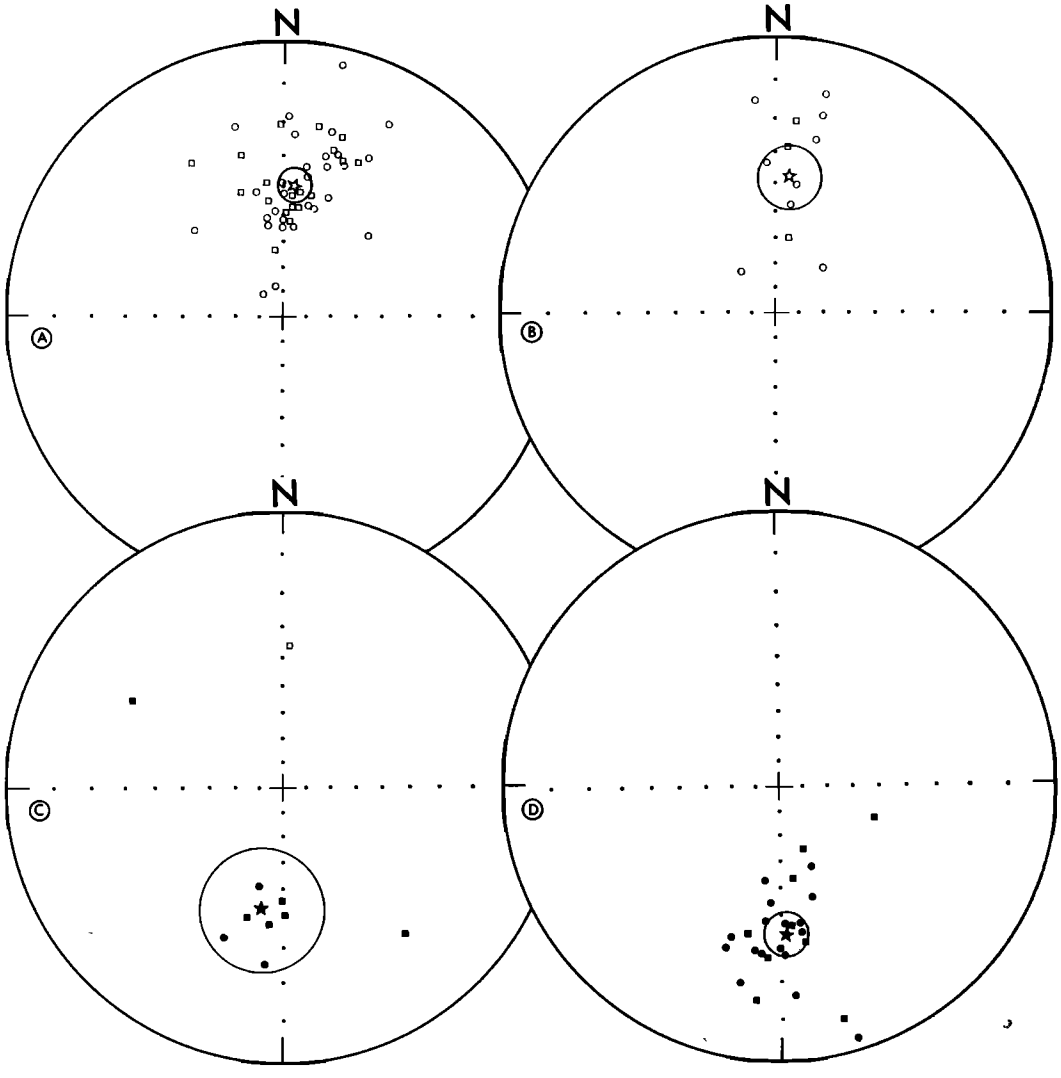


Fig. 5. Stereographic projections of the directions of magnetization with respect to the present horizontal. (*a*) Group 1 data, (*b*) group 3 normal polarities, (*c*) group 2 data, (*d*) group 3 reversed polarities. The squares represent the actual dated flows.

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magnetic directions. The second phenomenon, called dipole wobble, refers to the change in position of the average dipole axis with respect to the rotation axis. These two phenomena are usually collectively known as secular variation in the paleomagnetic literature, because they cannot be separated from one another using paleomagnetic data from only one locality. The third phenomenon is called polar wandering by which is meant the change of the rotation axis with respect to the sampling site. To be able to equate the paleomagnetically computed pole with the ancient rotation pole, it is necessary to assume that the scatter produced by the secular variation is essentially random in nature and that it can be averaged out by making enough observations over a long enough period of time. However, the question as to what constitutes an adequate time span and sampling density cannot be answered quantitatively until more is known about the amplitudes and time constants of the intrinsic variations of the geomagnetic field.

Since polar wandering rarely exceeds 1° per million years, this phenomenon can be ignored as a major source of scatter in the Réunion data. In the following paragraphs we will be mainly concerned with the data from group 1, because it contains the largest number of flows and is more nearly comparable with the Hawaiian results of *Doell and Cox* [1965]. Furthermore, as Figure 4 shows, the samples are fairly uniformly distributed throughout a well defined time interval of 600,000 years. The sampling density and time coverage of the data of groups 2 and 3 are not adequate for the analysis to be made in the following paragraphs; nevertheless, they are not inconsistent with the results of this analysis.

Dispersions due to nondipole components for Réunion. *Bullard et al.* [1950] have observed that the nondipole field drifts westward relative to the mantle at a rate of 0.18° per year. Thus, it may be reasonable to assume that deviations of the present geomagnetic field from an axial dipole field around a line of latitude at any time are the same as the variations that would occur at any point on that latitude over a suitable length of time. *Cox* [1962] and *Creer* [1962] suppose that the dispersion of the geomagnetic field directions at a given place during several tens of thousands of years may be estimated by rotating the present field about the geographic axis. An analysis of the field directions at intervals of 10° in longitude around a circle of latitude can then be made, using appropriate statistical methods. Such an analysis showed that the scatter to be expected is latitude-dependent and greater in the northern hemisphere than in the southern hemisphere. For the latitude of Réunion, Creer obtained $\delta_0 = 18.3°$, using the statistic

$$\delta_0 = 81/k^{1/2}$$

where k is the precision parameter for a Fisherian distribution of vectors [*Fisher*, 1953]. The corresponding δ_0 value for the Réunion data of group 1 is 19.8, which is not significantly different. Cox similarly analyzed the corresponding poles but used the present dipole axis as the true mean to obtain the best estimate of δ_0. Agreement with his analysis is less satisfactory if it is assumed that either the present dipole axis or rotation axis is the true mean of the Réunion data. However, neither of these two is probably the true mean. In Figure 6*a* the field directions for every 10° of longitude at latitude 20°S are plotted, using the field values tabulated by *Vestine et al.* [1947]. This nondipole curve would be comparable with a paleomagnetic collection of flows at 50-year intervals for about 2000 years. The actual collection represents samples at 10,000-year intervals and over a much longer period of time. In Figure 6*b* the Réunion nondipole curve is superimposed on the Réunion data of group 1, and this clearly shows that the area covered is the same in both cases. Therefore, the dispersion of the paleomagnetic data may be explained as being almost entirely due to variations in the nondipole field. Alternatively, one may conclude that the amplitude of the nondipole variations has remained nearly constant at 10°–15° over the past 600,000 years. Figure 6 also shows quite clearly that the area of maximum density of the paleomagnetic directions almost precisely coincides with the area of the nondipole curve. The closer spaced points on the nondipole curve correspond to observations in the Indian Ocean and the Pacific Ocean, where the nondipole field changes only slowly with longitude and the more widely spaced points correspond to the large anomaly

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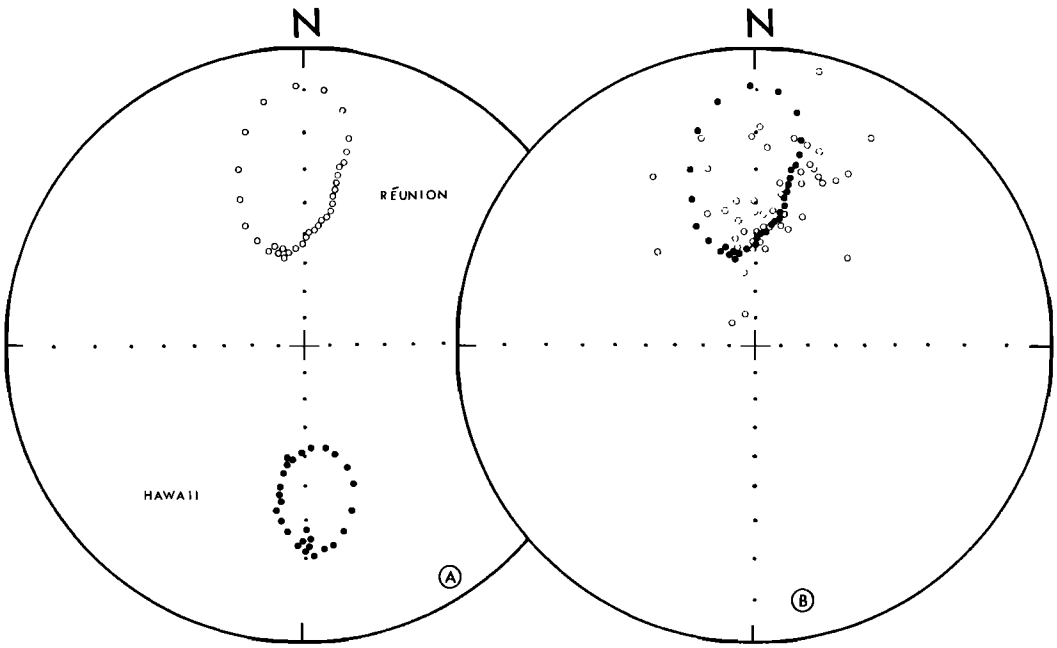


Fig. 6. (*a*) Nondipole curves for Réunion and Hawaii. (*b*) Réunion nondipole curve superimposed the Réunion group 1 data.

over southern Africa. Because this anomaly occupies only a small part of the line of latitude, the probability of it being recorded in the paleomagnetic data is small. Hence, the agreement between the two density distributions enhances the suggestion that the dispersion is due to the nondipole variations.

Comparison of the Réunion and Hawaiian dispersions. In Figure 6*a* the same nondipole curve appropriate to Hawaii has been plotted, and it clearly covers a much smaller area, illustrating the asymmetry between the northern and southern hemisphere. From their analysis of the Hawaiian paleomagnetic data *Doell and Cox* [1965] suggested that the low secular variation pattern in Hawaii had persisted for several tens of thousands of years. Since Réunion and Hawaii lie on the same latitude, the effects of dipole wobble and polar wandering should be roughly the same, and the difference in the dispersions should reflect the difference in the contribution of the nondipole contributions between the northern and southern hemisphere. This suggests that the asymmetry between the northern and southern hemisphere may well have persisted for 0.6 m.y. In reaching this conclusion, however, it has been assumed that the time span sampled and the sampling densities of both collections are comparable. For Réunion the age distribution is well documented, but for Hawaii the evidence is less clear. Radiometric dating has not been successful [*Evernden et al.*, 1964; *McDougall*, 1964] because of the low potassium content and high atmospheric contamination of the samples. As *Doell and Cox* [1965] have pointed out, the best estimate is given by *McDougall* [1964], who states that the age of the oldest volcanic unit is probably considerably less than 800,000 years. As no reversely magnetized rocks were found, the age is less than 700,000 years, because at this time the last reversal took place. It is therefore probably not unreasonable to assume that the time spans of both collections are the same. However, the possibility that the Hawaiian collection is based on five narrow intervals corresponding to the five volcanic units sampled by Doell and Cox cannot be excluded. Since Doell and Cox estimated that, on the basis of an average of 1000-year intervals between successive flows, the longest section represents only 50,000 years. If this estimate is true, the conclusion on the persistence of the asymmetry between the two hemi-

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spheres may well have to be modified as far as the duration is concerned. More detailed radiometric dating of the Hawaiian volcanics are necessary to resolve this difficulty.

Dipole wobble. The mean pole of the Réunion data of group 1 is in better agreement with the rotation pole than with the present inclined dipole. This suggests that the orientation of the inclined dipole is a purely temporary phenomenon.

It is clear from Figure 7, however, that the poles are not uniformly distributed around the geographic pole but form an elongated distribution. A chi-square test on the distribution [*Watson and Irving*, 1957] shows that it is not Fisherian. The mean pole and the geographic pole are 7° apart, and the circle of confidence does not include the geographic pole. Because these poles are not distributed in a Fisherian manner, it is not possible to test whether this difference is statistically different.

If the distribution of directions is governed primarily by the variations in the nondipole field, it follows that the distribution of the virtual paleomagnetic poles reflects the same variations. This implies, however, that the dipole wobble is a minor effect in comparison to the nondipole variations, and therefore it must have a time constant greater than 0.6 million years.

Although the Hawaiian poles are somewhat more uniformly distributed around the rotation pole, *Cox and Doell* [1964] point out that the distribution is also not Fisherian and furthermore both the Réunion and Hawaiian data have very few poles in the 0° to 90° quadrant, whereas in the other quadrants there is a fair degree of overlap. This observation can be interpreted as supporting evidence for the suggestion that the period of the dipole wobble is longer than 0.6 m.y.

The mean directions of Réunion and Hawaii are 10° apart. Unfortunately, the statistical procedure described by *Watson and Irving* [1957] cannot be applied because the two

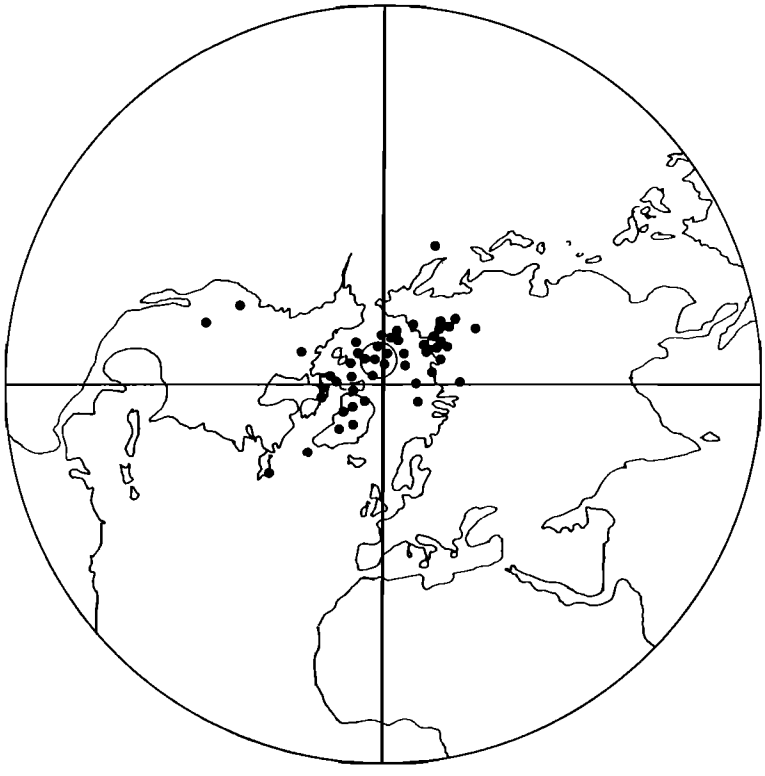


Fig. 7. Distribution of virtual paleomagnetic poles of the Réunion data for group 1.

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dispersions were found to be significantly different, nor can the fact that the two circles of confidence do not overlap be used because, as *Doell and Cox* [1965] have pointed out, these values are to be regarded as lower limits to the true confidence limits. The difference is, however, probably real and reflects the difference in the age distribution of the two collections.

Appendix

The flow average data after af demagnetization are given in the following table.

Sample	Mean Direction					Corresponding Pole	
	D, deg	I, deg	N	R	δ_{o}, deg	ϕ_{o}, deg	λ, deg
			Group 1				
RU70	022.7	−29.1	2	1.9932	3.33	134.6	67.7
RU71	350.0	−51.3	2	1.9944	2.9	271.5	75.8
RU72	348.2	−41.0	2	1.9995	0.7	309.4	78.8
RU75	359.3	−55.8	4	3.7146	10.8	236.8	74.5
RU76	011.8	−47.6	2	1.9982	1.6	183.2	76.7
RU77	017.0	−31.7	2	1.9991	1.0	134.7	73.4
RU78	004.6	−24.0	3	2.9947	1.9	083.9	80.4
RUDUM1	345.6	−29.8	2	1.7585	20.1	307.2	75.5
RU86	351.1	−64.5	2	1.9941	3.0	248.7	63.6
RU87	014.2	−42.5	2	1.9978	1.8	163.0	76.3
RU88	351.2	−42.7	2	1.9910	3.8	297.7	81.1
RU89	314.7	−44.6	2	1.9852	4.9	309.3	48.4
RUDUM2	007.8	−42.6	2	1.9884	4.3	173.8	81.8
RU32	011.0	−37.0	2	1.9755	6.3	144.4	79.7
RU33	350.5	−54.0	4	2.8343	22.5	264.2	74.0
RU34	004.1	−48.1	3	2.9372	6.7	211.0	80.9
RU35	355.6	−49.1	2	1.9992	1.0	257.2	80.1
RU36	011.0	−21.3	2	1.9939	3.1	103.7	75.4
RU37	009.1	−34.1	2	1.9937	3.1	132.1	81.0
RUDUM3	015.8	−21.1	2	1.9925	1.9	114.1	71.7
RU41	352.1	−44.3	2	1.9989	1.2	288.0	81.2
RU42	029.2	−13.5	1	.....	....	122.7	58.3
RUDUM4	005.2	−46.0	2	1.9768	6.1	199.2	81.9
RU47	001.3	−18.8	2	1.9991	1.1	062.0	78.6
RUDUM5	340.5	−39.2	2	1.9975	1.9	317.4	71.8
RU51	346.0	−77.8	3	2.3988	21.3	242.4	43.3
RU52	006.7	−54.0	2	1.9993	0.9	212.7	75.1
RU53	004.0	−52.7	3	2.9997	0.3	219.4	77.1
RUDUM6	345.2	−17.3	2	1.8902	13.4	286.7	71.3
RUDUM7	019.7	−26.2	2	1.9922	3.5	127.5	69.8
RUDUM8	329.0	−24.7	2	1.9967	2.5	314.8	59.4
RU68	359.5	−20.6	2	1.9941	3.0	237.7	79.6
RU82	003.7	−48.5	2	1.9994	0.8	213.9	80.7
RU84	046.6	−44.8	2	1.9998	0.3	161.7	47.1
RU85	005.0	−41.5	2	1.9979	1.8	177.2	84.4
RU91	013.0	−04.0	2	1.9979	1.7	090.6	67.1
RUDUM9	018.1	−26.5	3	2.9191	7.6	125.9	71.3
RU94	028.0	−24.8	2	1.9752	6.3	133.0	62.1
RU103	320.6	−78.5	2	1.5759	26.0	252.4	37.8
RU104	015.0	−45.6	2	1.9979	1.7	172.0	75.0
RU105	019.7	−22.7	1	.....	....	122.4	68.9
RU106	025.2	−26.6	2	1.9968	2.2	133.2	64.9
RU107	020.7	−40.3	2	1.9986	1.4	154.9	70.6
RU108	015.8	−28.3	2	1.9992	1.0	126.3	73.7
RU116	023.6	−29.6	2	1.9991	1.1	136.2	66.9
RU121	000.2	−42.0	2	1.9967	2.2	230.1	86.7
RU122	020.5	−29.1	2	1.9994	0.8	132.9	69.8
RU123	016.8	−45.2	3	2.9994	0.5	168.8	73.5
RU124	351.0	−40.2	6	5.9807	1.8	310.0	81.4

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Sample	Mean Direction					Corresponding Pole	
	D, deg	I, deg	N	R	δ_0	ϕ_0, deg	λ, deg
			Group 2 Normal				
RUDUM11	003.0	−35.8	3	2.9956	1.7	124.7	86.6
			Group 2 Reversed				
RUDUM10	179.0	43.2	2	1.9991	1.0	67.0	−85.6
RU13	187.2	38.0	3	2.9605	5.3	329.7	−83.1
RU61	194.6	50.0	2	1.9839	5.1	004.5	−73.5
RU62	139.3	21.5	2	1.9991	1.0	152.6	−49.9
RU63	181.8	46.3	2	1.9984	1.5	040.9	−83.0
RU64	196.0	38.3	3	2.9972	1.3	330.2	−75.0
RU65	300.0	25.5	2	1.9962	2.4	300.0	21.8
RU66	202.8	30.8	2	1.9982	1.6	317.5	−67.9
RU67	186.7	25.5	2	1.9981	1.7	276.0	−79.8
			Group 3 Normal				
RUDUM12	011.8	−60.0	2	1.8981	12.9	210.9	67.5
RU3	357.5	−33.6	2	1.9929	3.3	278.0	86.5
RUDUM13	005.0	−29.0	3	2.8575	10.2	099.0	82.9
RU144	355.6	−15.5	1	.....	....	253.4	76.4
RU145	007.6	−22.6	1	.....	....	094.9	78.2
RU135	013.1	−11.6	1	.....	....	097.1	70.2
RU30	319.6	−68.7	2	1.9992	1.0	269.8	46.0
RU161	014.8	−25.0	1	.....	....	118.5	73.7
RU162	014.6	−18.1	1	.....	....	107.7	71.6
RUDUM20	009.1	−47.2	3	2.9872	3.0	188.5	78.8
RU76	010.6	−40.3	1	.....	....	158.6	79.8
RU177	046.5	−63.2	1	.....	....	188.2	45.1
			Group 3 Reversed				
RUDUM14	165.3	08.6	3	2.8393	10.8	192.5	−68.1
RUDUM15	185.0	27.5	2	1.9927	3.3	272.3	−82.0
RU12	179.6	27.7	2	1.9973	2.0	232.3	−83.7
RU132	178.3	37.0	1	.....	....	158.0	−88.4
RU134	171.0	53.1	1	.....	....	085.0	−74.9
RU136	193.0	33.1	1	.....	....	314.1	−77.3
RU137	172.5	39.0	1	.....	....	135.0	−83.0
RU139	186.8	27.0	1	.....	....	280.5	−80.6
RU140	189.7	27.6	1	.....	....	292.5	−78.7
RU141	172.7	35.1	1	.....	....	156.8	−83.5
RU142	159.1	55.5	1	.....	....	100.6	−66.3
RU147	189.3	52.5	1	.....	....	022.5	−75.2
RU148	159.0	63.7	1	.....	....	085.2	−60.0
RU13	187.2	38.0	3	2.9605	5.3	329.7	−83.1
RU14	163.6	03.5	2	1.9986	1.4	193.2	−65.0
RU16	185.5	44.7	3	2.9992	0.6	013.1	−82.6
RU17	177.8	37.0	2	1.9951	2.7	154.8	−88.0
RU18	163.2	45.2	3	2.9877	2.9	120.9	−73.6
RUDUM18	107.6	31.7	8	6.9465	10.8	166.6	−80.4
RUDUM16	200.6	27.5	4	3.9711	3.4	310.3	−69.2
RU27	187.1	15.3	3	2.9914	2.4	263.9	−75.1
RU28	192.0	18.6	2	1.9936	3.2	282.1	−73.7
RUDUM17	176.0	16.5	3	2.9542	5.7	217.7	−76.9
RU158	199.1	30.7	1	.....	....	314.6	−71.3
RU160	181.8	29.2	1	.....	....	254.0	−84.3
RU179	109.2	50.1	3	2.9949	1.8	119.9	−26.6

Notes

N, number of specimens per sample, except when the sample name is RUDUM. It then indicates the number of samples per flow.

ϕ, λ, longitude and latitude of the pole.

R, Resultant vector length.

δ_0, angular dispersion factor.

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